Complete Chapter 5 Practice Assessment

1. Determine the following limits.
   (a) $\lim_{x\to-\infty} -2x^4 =$
   (b) $\lim_{x\to\infty} 2x^{-4} =$
   (c) $\lim_{x\to-\infty} 5\cdot\frac{3}{x} =$
   (d) $\lim_{x\to-\infty} -2x^4 =$

2. Write the function $\frac{-3(x^2)^4}{5}$ as a power function of the form kx^p. Enter the values of k and p in the spaces provided.

   $k =$ _____ $p =$ _____

3. Circle all of the polynomials.
   (a) $f(x) = 6^x + 4$
   (b) $f(x) = x^6 + 4$
   (c) $f(x) = 6x^{3.7} + 2$
   (d) $f(x) = 6x^2 + 3.7$
   (e) $f(x) = 2x^{-6} + 3.7$

4. Determine the degree and the long-run behavior of each polynomial.
   (a) $f(x) = 9x^6 - 3961x^2 + 6$

   degree of f : _____ $\lim_{x\to\infty} f(x) =$ _____ $\lim_{x\to-\infty} f(x) =$ _____

   (b) $f(x) = 8x^3 + \frac{6x^4}{x^{-6}} - 7x^5 + 1$

   degree of f : _____ $\lim_{x\to\infty} f(x) =$ _____ $\lim_{x\to-\infty} f(x) =$ _____

5. Determine the zeros of the following polynomials. Do not approximate.
   (a) $f(x) = x^3(7 - 2x^9)$
   (b) $f(x) = (5 - 2x)(3x - 2)^2$

6. An open-top box is to be constructed from a 6 inch by 14 inch rectangular sheet of tin by cutting out squares of equal size at each corner, then folding up the resulting flaps. Let x denote the length of the side of each cut-out square. Assume negligible thickness.
   (a) Find a formula for the volume, V, of the box as a function of x.
   (b) For what values of x does your formula make sense in the context of the problem?
   (c) Create a sign chart and sketch the graph of $V(x)$.

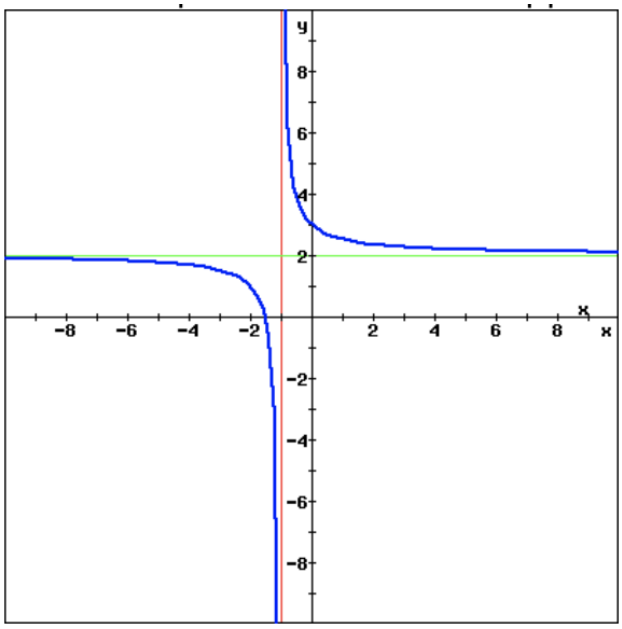
7. Find the horizontal asymptote, if it exists, of the rational function below.

$$g(x) = \frac{(-1-x)(-7-2x)}{2x^2+1}$$

8. Find all zeros and vertical asymptotes of the rational function below.

$$f(x) = \frac{x^2 - 16}{-x^3 - 16x^2}$$

9. The graph below is a vertical or horizontal shift of $y = \dfrac{1}{x}$ so that the horizontal asymptote is $y = 2$ and the vertical asymptote is $x = -1$ (assume no reflections or compression/expansions have been applied).



(a) The graph's equation can be written in the form

$$f(x) = \frac{1}{x + A} + B.$$

Determine A and B.

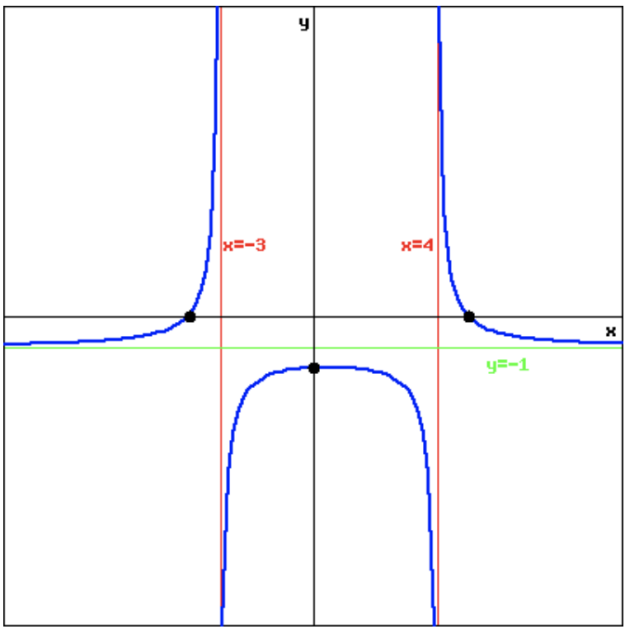
(b) Your formula from part (a) can be rewritten it as a ratio of linear polynomials,

$$f(x) = \frac{Mx + C}{x + D}.$$

Determine M, C, and D.

(c) Compute the exact values of the x and y intercepts.

10. Find a possible formula for the function graphed below.The x intercepts are 5 and -4, the y-intercept is $-\dfrac{5}{3}$, the horizontal asymptote is $y = -1$, and the vertical asymptotes are $x = -3$ and $x = 4$. Give your formula as a reduced rational function.


y
x
x=-3
x=4
y=-1